CAGE-Fake: Article Trustworthiness Audit Report
Calibration-Aware Gating & Evidence-Grounded Reasoning for Trustworthy Fake News Triage

AUDIT ID: AUD-247238
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FRAMEWORK: CAGE-Fake v2.0

LEGAL & FACT-CHECKING POLICY: A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify factual reality. URL extraction, pattern recognition, and external evidence verification remain strictly separate processes. The pipeline abstains (**UNCERTAIN**) whenever evidence is missing, conflicting, or model consensus is low.

Title: Man gets life term for abducting, raping minor
Source: https://timesofindia.indiatimes.com/city/agra/man-gets-life-te...
Word Count: 215 words | **Extraction:** Successfully Extracted via Trafilatura
OOD Detector: Loaded (SBERT & Vocab) | **Triage Policy:** Evidence-Grounded Three-Way Gating

TRIAGE VERDICT: UNCERTAIN
Abstention Confidence: 50.0% (τ = 80.0%)
Model Disagreement (std): 0.208 (δ = 0.15)
Safety Trigger: Evidence Absence Abstain (Unverified Topic)

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Vote	Confidence
Random Forest	42.1%	REAL	57.9%
XGBoost	93.9%	FAKE	93.9%
LightGBM	91.1%	FAKE	91.1%
Logistic Regression	69.5%	FAKE	69.5%
CAGE-Fake Stack	88.8%	FAKE	88.8%

Consensus Status: **High Dissent / Conflicting Predictions**
Vote Distribution: **4 FAKE** vs **1 REAL**
Base Disagreement (4-base std): 0.208 (δ = 0.15)
5-Model Consensus Spread: 0.195
[==========------]
Safety Assessment: Base-model disagreement (0.208) exceeds safety threshold delta (0.150).

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	90.3/100	Excellent / Restrained
Source Attribution	75.0/100	Adequate / Neutral
Epistemic Grounding	90.3/100	Excellent / Restrained
Punctuation Discipline	88.2/100	Excellent / Restrained
Readability Structure	90.3/100	Excellent / Restrained
Lexical Diversity	79.0/100	Adequate / Neutral

OVERALL CREDIBILITY INDEX
85.1 / 100
Credibility Tier: **A (Restrained / Professional Copy)**

Scores are standardized against verified newsroom copy standards: high factual attribution, balanced syntax, and low sensationalism maximize the index.

3. Grounded Diagnostic Reasoning & Counterfactual Sensitivity

Executive Synthesis: System abstains with UNCERTAIN triage (4/4 claims (100%) returned no live corroborating sources. Style model predicts FAKE (88.8%) but this signal cannot be trusted without factual grounding — the vocabulary may simply be absent from the training distribution (domain shift). System abstains; manual editorial review required.). Multi-view divergence (std=0.208) mandates human editorial review.

Diagnostic Explanation: 1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **50.0% calibrated confidence**.
- **Gating Trigger**: System abstention triggered under safety policy (evidence_absence_abstain).

Counterfactual Simulation: Neutralizing 1 linguistic markers resulted in a signed probability shift of +1.6 percentage points (from 88.8% to 90.3%), retaining the text classifier's verdict (FAKE) (note: the final triage verdict may differ when evidence overrides it). Original P(FAKE): **88.8%** → Transformed P(FAKE): **90.3%** **(Signed Shift: +1.6%)**

[v] Clean Stylometrics: No significant sensational exaggeration, clickbait patterns, or manipulative phrasing detected.

4. Claim-Level Semantic Evidence Grounding (Triage Verification)

Extracted Factual Claim	Verification Status	Authoritative Source / Match	Evidence Context Quote
Agra: A special Pocso court in Agra has sentenced a 30-year-old man to life impris...	UNVERIFIED (Sim: 0.49)	No live sources >= 0.60 similarity	No corroborating external sources met the >= 0.60 semantic similarity floor.
During the trial, the prosecution examined 12 witnesses, including the complainant...	UNVERIFIED (Sim: 0.52)	No live sources >= 0.60 similarity	No corroborating external sources met the >= 0.60 semantic similarity floor.
While delivering the verdict, special judge (Pocso) Sonika Chaudhary highlighted s...	UNVERIFIED (Sim: 0.68)	indiatoday.in	Sonika Chaudhary, Special Judge of POCSO court, upon reviewing
Man gets life term for abducting, raping minor.	UNVERIFIED (Sim: 0.51)	No live sources >= 0.60 similarity	No corroborating external sources met the >= 0.60 semantic similarity floor.

5. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Impact	Top REAL-Driving Anchors	Impact
LR [TF-IDF Lexical]	+1.444	RF [Stylometrics]	-0.223
XGB [SBERT Semantic]	+0.855	Inter-View Disagreement (std)	-0.208
LightGBM [SVD Topics]	+0.574	-	-

6. Empirical Benchmark Model Comparison (Held-out Test Split)

Model Name	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest (Uncalibrated)	0.8098	0.8097	0.0770	100.0%	0.8098
XGBoost (Uncalibrated)	0.8508	0.8505	0.0550	100.0%	0.8508
LightGBM (Uncalibrated)	0.8787	0.8785	0.0401	100.0%	0.8787
Logistic Regression (Uncalibrated)	0.9361	0.9361	0.1322	100.0%	0.9361
Uncalibrated Stack	0.9410	0.9410	0.0182	100.0%	0.9410
Calibrated Stack (Full Coverage)	0.9410	0.9409	0.0274	100.0%	0.9410
CAGE-Fake (Calibrated + Gated)	**0.9410**	**0.9409**	**0.0274**	**82.5%**	**0.9781**

7. Research Calibration & Disagreement Visualizations







